WESTMINSTER
International University in Tashkent

System Analysis & Design

Tutorial 05

Agenda

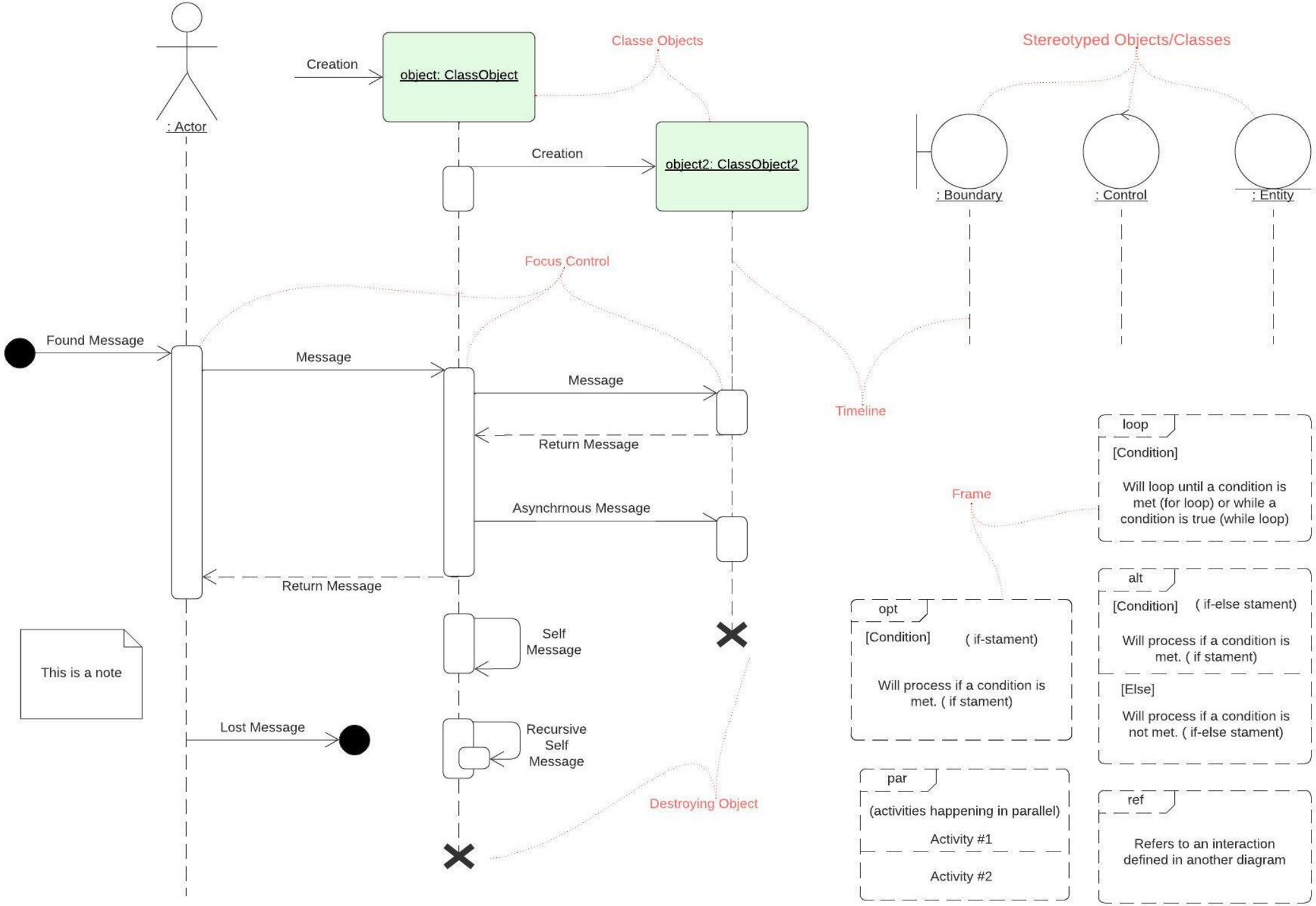
WIUT

1. Previous tutorial recap
    1. Sequence diagrams
2. Requirements Gathering

Previous tutorial recap

WIUT

1. Sequence diagrams
   - What does it record?
   - What shapes does it have?


: Actor
Creation
object: ClassObject
Classe Objects
Creation
object2: ClassObject2
Focus Control
Found Message
Message
Message
Return Message
Asynchrnous Message
Return Message
Self
Message
This is a note
Lost Message
Recursive
Self
Message
Destroying Object
Timeline
Stereotyped Objects/Classes
: Boundary
: Control
: Entity
Frame
loop
[Condition]
Will loop until a condition is met (for loop) or while a condition is true (while loop)
opt
[Condition] (if-stament)
Will process if a condition is met. (if stament)
alt
[Condition] (if-else stament)
Will process if a condition is met. (if stament)
[Else]
Will process if a condition is not met. (if-else stament)
par
(activities happening in parallel)
Activity #1
Activity #2
ref
Refers to an interaction defined in another diagram

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Requirements Gathering

How to elicit what business really needs?

Requirement Gathering

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What requirements gathering techniques can you think of?

Requirement Gathering

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What requirements gathering techniques can you think of?

- Interview
- Survey / Questionaire
- Read existing documentation
- Observation (watch users perform their duties)
- Prototyping
- Brainstorming
- Focus groups / Workshops
- Join Application Design (JAD)

Exercise: Identify the method

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When to use:

1. When users struggle to express needs verbally.
2. When there are many stakeholders or large user groups.
3. When detailed, qualitative insights are needed.
4. When actual work practices differ from documented ones.

Advantage:

1. Provides visual feedback; reduces ambiguity.
2. Quick data collection from many people; easy to analyze.
3. Builds relationships; allows follow-up questions.
4. Reveals real processes; uncovers hidden needs.

Disadvantage:

1. Time-consuming; can raise unrealistic expectations.
2. Limited depth; unclear questions may confuse respondents.
3. Time-consuming; may be biased or incomplete.
4. Can be intrusive; users may change behavior when observed.

Requirement Gathering

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- There are many stakeholders in every system.
- They can have different perspectives.
- Your job is to collect requirements that are:
  - Complete
  - Accurate
  - Unambiguous
- No single method fits all cases. You should combine several techniques.

Quality Requirements

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Don’t forget about the techniques we mentioned previously:

- Five Whys
- Prototyping
- Involving the end users early on

Exercise: Group

1. Imagine you are asked to develop SRS system for the university.
2. Who are the stakeholders?
3. What type of requirements gathering techniques would you use & why?
4. Who would you contact to gather the requirements?
5. What could be 5 functional and 3 non-functional requirements you would document?

Exercise: Role-Play

1. Divide into groups of 3.
2. Each group has 3 roles:
   - System Analyst
   - Stakeholder (Registrar, student, teacher, IT, etc.)
   - Observer/Note-taker.

3. System analyst interviews the “client” for 10 minutes.
4. Observer takes notes about the communication style and clarity.
5. Each group will **share 3 functional, 1 non-functional requirement and a challenge** they faced during the process.

Exercise: Role-Play

1. Each group **shares 3 functional, 1 non-functional requirement and a challenge** they faced during the process.
2. What questions worked best to uncover important requirements?
3. Did you use any techniques? If yes, what was the result?
4. How are you sure you understood the stakeholders' statements correctly?



The End

See you at the lecture.